

Assignment No. 6

Experiment Title: Exploring Tableau

Problem Statement:

Explore interfaces of Tableau

Objective:

1. Understanding Tableau Interface Navigation.
2. Exploring Interactive Features and Customization Options.

Theory:

1. Write a brief description of Tableau software.

Tableau is a powerful and fastest-growing data visualization tool used in the Business Intelligence Industry. It is a leading data visualization tool used for data analysis and business intelligence. Tableau is an American company that started in 2003, and in June 2019, Salesforce acquired Tableau. It helps users create different charts, graphs, maps, dashboards, and stories for visualizing and analysing data to aid in business decisions. Tableau has a lot of unique, exciting features that make it one of the most popular tools in business intelligence (BI).

Tableau supports powerful data discovery and exploration, enabling users to answer important questions in seconds. No prior programming knowledge is needed; users without relevant experience can start immediately with creating visualizations using Tableau. It can connect to several data sources that other BI tools do not support. Tableau enables users to create reports by joining and blending different datasets. Tableau Server supports a centralized location to manage all published data sources within an organization.

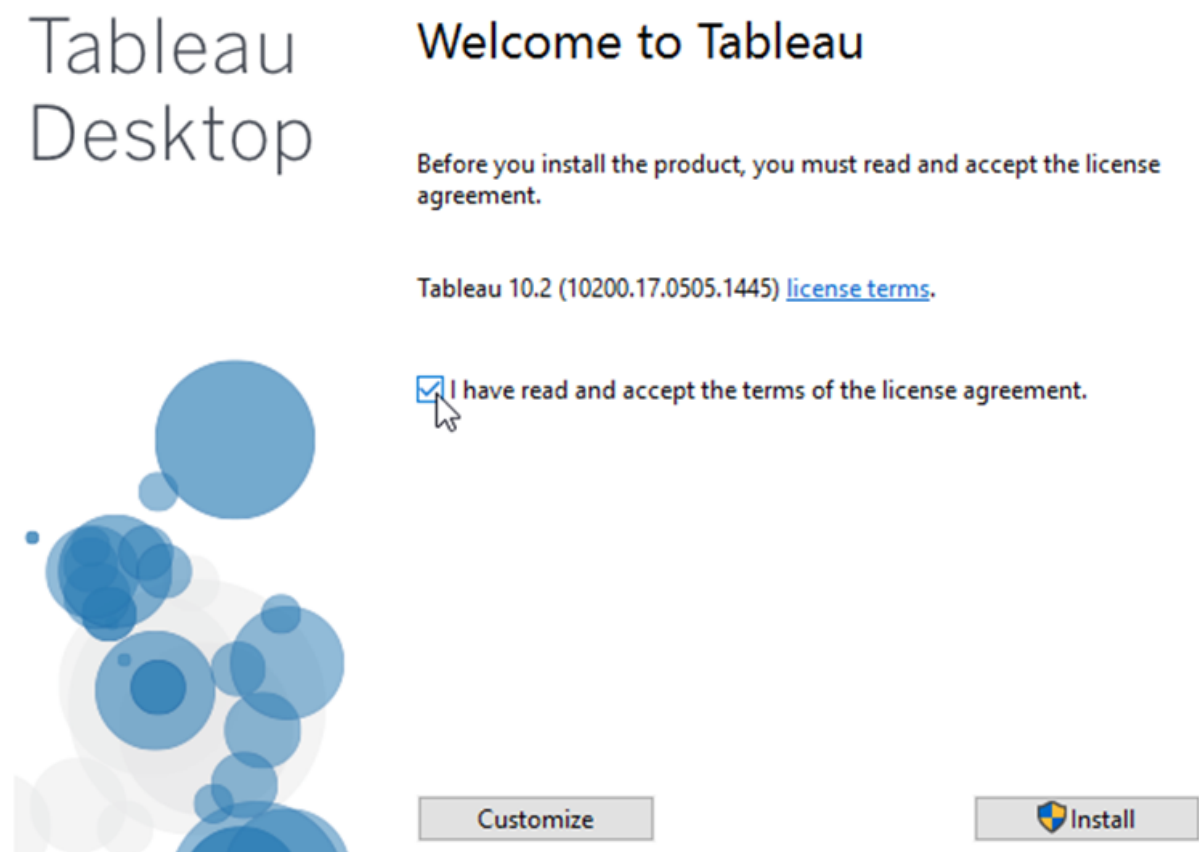
Tableau offers various products, including Tableau Desktop, Tableau Server, Tableau Online, Tableau Public, and Tableau Reader. Tableau Desktop is used to create reports, dashboards, and stories using different charts and graphs, with workbooks and dashboards that can be shared locally or publicly. Tableau Server is mainly used in organizations to share workbooks and reports that are created using Tableau. Tableau Online is a cloud-based platform that helps users publish dashboards and share discoveries with anyone, connecting to cloud databases like Amazon Redshift and Google BigQuery. Tableau Public is free to use, but the workbooks created cannot be saved locally, and they can be saved to Tableau's public cloud, which anyone can access

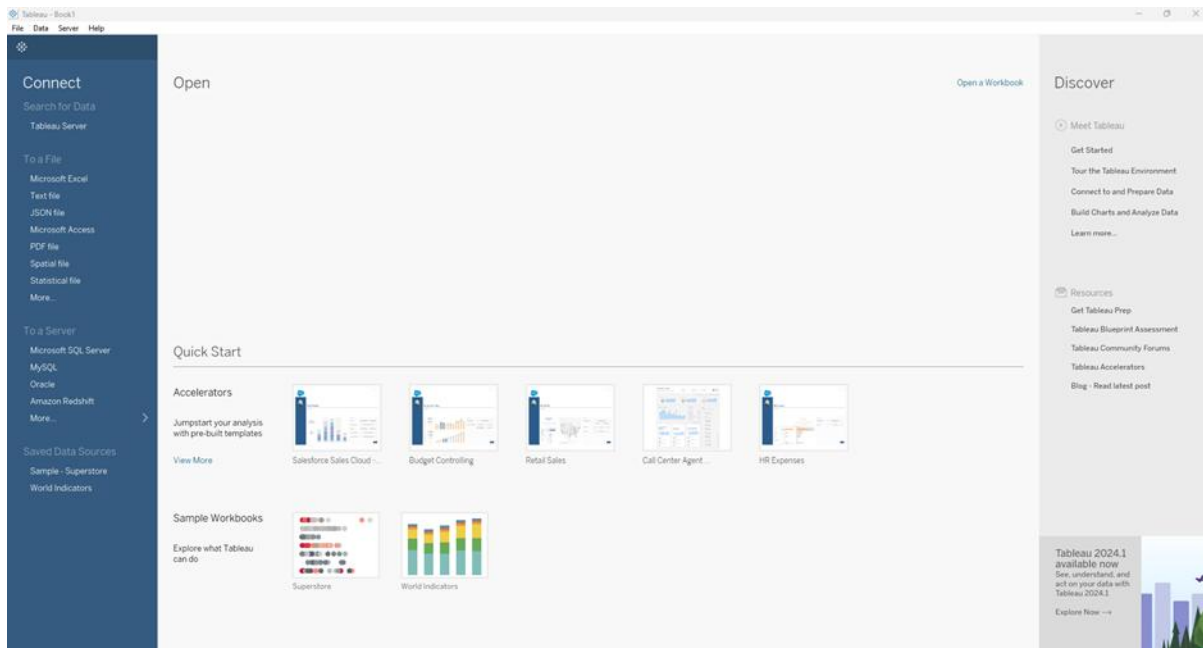
and view. Tableau Reader is a free desktop application that enables users to open and interact with reports built using Tableau Desktop.

Tableau has a user-friendly interface, making it easy for non-technical analysts and end-users to convert data into understandable ones. It provides features like cleaning, organizing, and visualizing data, making it easier to create interactive visual analytics in the form of dashboards. Tableau dashboards are very interactive and easy to draw, with no manual calculation required, as all the calculations are done by the tableau only. However, Tableau is a paid tool, which can be a disadvantage for some users.

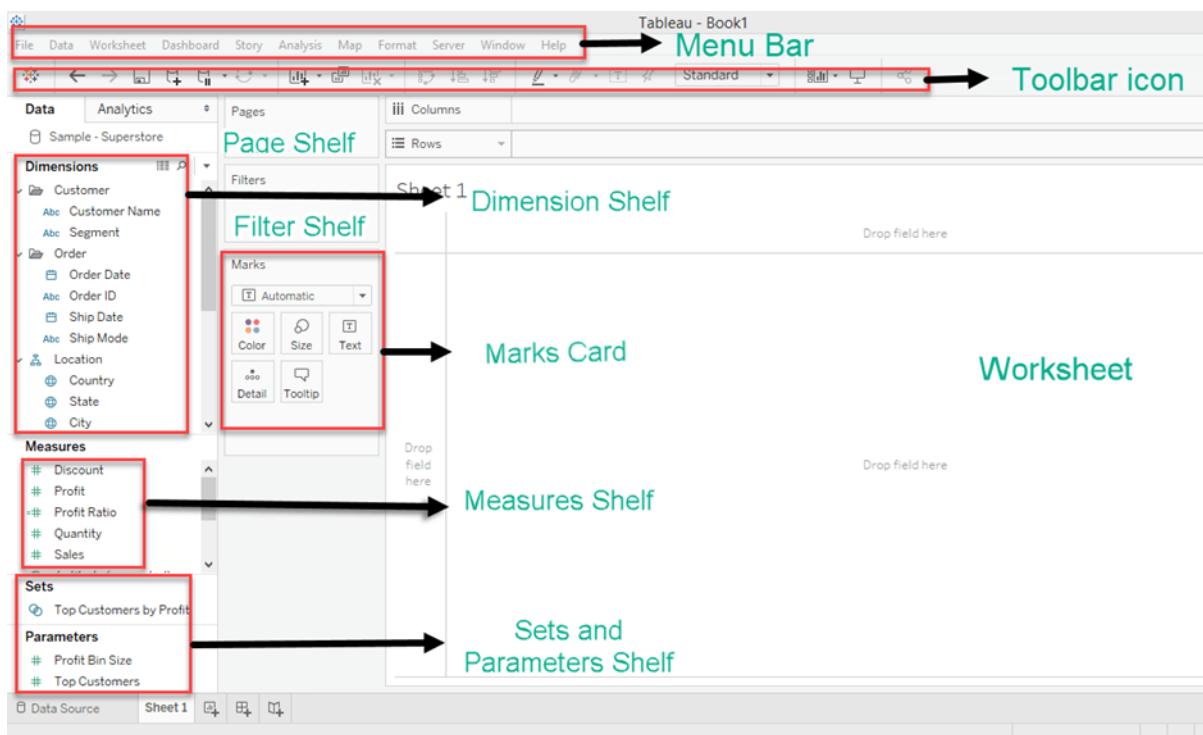
2. With screenshots , explain the installation of Tableau software.

<https://www.tableau.com/products/desktop>





3. With suitable diagrams explain the components in Tableau software.



The interface of Tableau Desktop software consists of various components, including:

- **Menu Bar:** The menu bar is a strip of options found at the top of the Tableau Desktop window, which provides a set of options to help users find information or execute functions.
- **Toolbar:** The toolbar is a strip of icons that can be selected to perform certain functions, such as the story toolbar.

- **Shelves:** Shelves are Tableau Desktop-specific interface components where users can drag and drop fields (or “pills”) to organize and filter data in their visualizations. Examples include the Filters shelf and the Pages shelf.
- **Cards:** Cards are Tableau Desktop-specific interface components that give detailed information about the view and enable interactivity. Examples include the Marks card, a Filter card, a Slider card, or a Colour Legend card.
- **Content Area:** The content area is the text input area used to edit tooltip text and title text.
- **Data Pane:** The Data Pane in Tableau Desktop allows users to connect and select data sources for their visualizations, providing a list of tables and columns from the data source.
- **Dimensions Pane:** The Dimensions Pane in Tableau Desktop contains categorical data, such as names, dates, and geographic locations, which can be used to create visualizations that show trends, patterns, and comparisons across different categories.
- **Measures Pane:** The Measures Pane in Tableau Desktop contains quantitative data, such as numbers, percentages, and ratios, which can be used to create visualizations that show aggregations, calculations, and trends across different data points.
- **Marks Card:** The Marks Card in Tableau Desktop allows users to customize the appearance and behavior of visualizations by selecting different mark types, such as bars, lines, circles, and shapes. Users can also apply formatting, colors, and labels to the marks.
- **Dialog Box:** A dialog box is a clickable area that shows another page/area of the application, providing additional options or settings for the user.

Conclusion: In conclusion, mastering Tableau's interface and interactive features is key for creating impactful data visualizations. Understanding navigation and customization options empowers users to extract valuable insights from data and effectively communicate findings. With proficiency in Tableau, users can drive informed decision-making and enhance business outcomes through compelling data-driven narratives